Content Xref Feed — Anomaly Detection Strategy Document

Overview

This document provides a structured anomaly detection strategy for the Content Xref Feed using AWS Glue Data Quality anomaly detection metrics.

The primary objective of anomaly detection is to identify unusual, invalid, incomplete, inconsistent, or risky data patterns before they impact downstream analytics, marketing operations, regulatory reporting, campaign execution, or enterprise business processes.

The Content Xref Feed contains document metadata, product mappings, audience information, content classifications, platform information, lifecycle dates, marketing content attributes, and delivery metadata. Because of this, anomaly detection is critical for ensuring content governance, operational stability, compliance, and business trust.

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1. Available AWS Anomaly Search Metrics

The following anomaly search metrics are officially supported by AWS Glue Data Quality and can be used for profiling, monitoring, and anomaly detection in enterprise data pipelines.

Metric	Description
ColumnLength	Measures the length characteristics of column values and helps identify abnormal text growth, corrupted payloads, truncated content, or malformed metadata. Useful for detecting oversized subject lines, corrupted tactic names, or abnormal URL lengths.
ColumnValues	Validates whether column values match expected or allowed business values. Helps identify invalid statuses, unsupported classifications, unexpected countries, or malformed categorical data entering the feed.
Completeness	Measures the percentage of non-null values in a column. Useful for detecting missing mandatory business fields, ingestion issues, or upstream ETL failures affecting content metadata.
Distinct ValuesCount	Measures the number of unique values within a column. Helps identify unexpected category growth, new business classifications, or abnormal reductions in data diversity.

Metric	Description
RowCount	Measures the total number of records processed within the dataset. Useful for detecting feed failures, duplicate ingestion, missing partitions, or unusual volume spikes.
Uniqueness	Measures whether values expected to remain unique continue to do so. Commonly used to detect duplicate document identifiers or repeated content records.
Mean	Calculates the average value for numeric fields. Useful for detecting statistical drift or abnormal changes in numerical business metrics over time.
Sum	Calculates total aggregated numeric values across records. Helps identify abnormal increases or decreases in business aggregates or feed-level measurements.
StandardDeviation	Measures the variability of numeric values relative to the average. Useful for detecting unstable distributions, inconsistent behavior, or abnormal fluctuations in business patterns.
Entropy	Measures distribution randomness or diversity of values in a column. Useful for detecting distribution drift such as abnormal concentration of countries, products, platforms, or content types.
Unique ValueRatio	Measures the ratio between unique values and total records. Useful for detecting duplicate spikes, repeated content reuse, or declining data uniqueness.
ColumnCount	Measures the total number of columns in the dataset. Helps identify schema drift, missing columns, unexpected new fields, or structural changes in the source feed.
ColumnCorrelation	Measures relationships between columns and identifies abnormal changes in expected business relationships. Useful for detecting incorrect mappings or inconsistent metadata relationships.
CustomSql	Enables implementation of custom business-rule validations using SQL logic. Useful for validating lifecycle rules, compliance logic, date consistency, and enterprise governance requirements.
AllStatistic	Generates a combined statistical profile across multiple metrics simultaneously. Useful for broad anomaly monitoring, enterprise observability, and automated data profiling.

2. Top 20 Business-Critical Anomalies for Content Xref Feed

#	Anomaly	AWS Metric
1	Sudden content feed row drop/spike	RowCount
2	Duplicate `document_id` increase	Uniqueness
3	NULL spike in `document_id`	Completeness
4	NULL spike in `product_apims_id`	Completeness
5	NULL spike in `country`	Completeness
6	Invalid `document_status` values appearing	ColumnValues
7	Invalid `content_type` values appearing	ColumnValues
8	Sudden increase in expired content	Entropy
9	Unexpected new `classification` values	Distinct ValuesCount
10	Unexpected new `platform` values	Distinct ValuesCount
11	Country distribution drift	Entropy
12	Product distribution drift	Entropy
13	Duplicate content ratio increase	Unique ValueRatio
14	Sudden drop in unique documents	Distinct ValuesCount
15	Missing `approved_job_id` spike	Completeness
16	Abnormal `subjectline` length increase	ColumnLength
17	Product and indication relationship drift	ColumnCorrelation
18	Country and language relationship drift	ColumnCorrelation
19	Unexpected schema/column changes	ColumnCount
20	Invalid content lifecycle (`expiry_date < approved_start_date`)	CustomSql

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3. Detailed Explanation of Each Recommended Anomaly

1. Sudden Content Feed Row Drop/Spike

Metric: RowCount

Detects abnormal increases or decreases in the total number of content records being ingested. This anomaly helps identify feed failures, delayed processing, duplicate ingestion, or upstream extraction issues.

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2. Duplicate `document_id` Increase

Metric: Uniqueness

Detects situations where document identifiers expected to remain unique begin appearing multiple times. This helps prevent duplicate reporting, repeated content usage, and downstream analytics corruption.

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3. NULL Spike in `document_id`

Metric: Completeness

Monitors missing values in the primary document identifier column. Missing document identifiers can break traceability, governance, and downstream joins.

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4. NULL Spike in `product_apims_id`

Metric: Completeness

Detects missing product mappings associated with content records. Product mappings are critical for commercial analytics, compliance, and reporting consistency.

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5. NULL Spike in `country`

Metric: Completeness

Identifies missing country information in content metadata. Country-level visibility is essential for regional governance, regulatory reporting, and operational monitoring.

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6. Invalid `document_status` Values Appearing

Metric: ColumnValues

Detects unsupported or unexpected document status values entering the feed. This helps identify workflow inconsistencies, source corruption, or upstream changes.

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7. Invalid `content_type` Values Appearing

Metric: ColumnValues

Monitors content type classifications for unsupported or abnormal values. This helps prevent downstream categorization issues and business-rule violations.

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8. Sudden Increase in Expired Content

Metric: Entropy

Detects abnormal increases in expired content records compared to historical behavior. This may indicate lifecycle management failures or delayed archival processes.

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9. Unexpected New `classification` Values

Metric: Distinct ValuesCount

Detects new classifications appearing in the feed that were not historically present. Useful for identifying taxonomy drift or uncontrolled business expansion.

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10. Unexpected New `platform` Values

Metric: Distinct ValuesCount

Monitors unexpected growth in platform diversity. Helps identify unsupported delivery platforms, source-system changes, or integration anomalies.

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11. Country Distribution Drift

Metric: Entropy

Detects abnormal shifts in content distribution across countries. This helps identify regional ingestion issues, missing data, or unusual concentration patterns.

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12. Product Distribution Drift

Metric: Entropy

Monitors abnormal changes in product-related content distributions. Useful for identifying mapping corruption, commercial imbalance, or upstream inconsistencies.

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13. Duplicate Content Ratio Increase

Metric: Unique ValueRatio

Detects increasing duplication across content records. This helps identify repeated ingestion, content reuse anomalies, or identity duplication problems.

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14. Sudden Drop in Unique Documents

Metric: Distinct ValuesCount

Detects abnormal reductions in the number of distinct content records. Often indicates missing data partitions, partial ingestion, or feed processing failures.

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15. Missing `approved_job_id` Spike

Metric: Completeness

Monitors completeness of approval workflow identifiers. Missing approval references can impact auditability, governance, and compliance tracking.

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16. Abnormal `subjectline` Length Increase

Metric: ColumnLength

Detects unusual growth in email subject line size. This may indicate corrupted payloads, HTML injection, malformed metadata, or formatting issues.

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17. Product and Indication Relationship Drift

Metric: ColumnCorrelation

Monitors changing relationships between products and disease indications. Unexpected changes may indicate mapping corruption or metadata inconsistencies.

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18. Country and Language Relationship Drift

Metric: ColumnCorrelation

Detects abnormal changes in expected relationships between countries and supported languages. Useful for identifying localization issues or incorrect regional mappings.

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19. Unexpected Schema/Column Changes

Metric: ColumnCount

Detects structural dataset changes such as missing columns or newly added fields. Schema anomalies can break downstream pipelines, validations, and analytics.

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20. Invalid Content Lifecycle (`expiry_date < approved_start_date`)

Metric: CustomSql

Uses custom SQL validation logic to detect invalid lifecycle date ranges. This helps ensure content validity and prevents logically incorrect records from entering downstream systems.

Conclusion

The recommended anomaly detection strategy for the Content Xref Feed focuses on business-critical data quality monitoring using AWS Glue Data Quality metrics.

The implementation should prioritize:

1. Feed health monitoring
2. Mandatory metadata completeness
3. Duplicate prevention

4. Lifecycle validation
5. Distribution drift monitoring
6. Relationship consistency
7. Schema stability
8. Governance and compliance validation

By implementing these anomaly checks, the organization can significantly improve content governance, data reliability, operational visibility, and downstream analytical trust across enterprise marketing and content management systems.